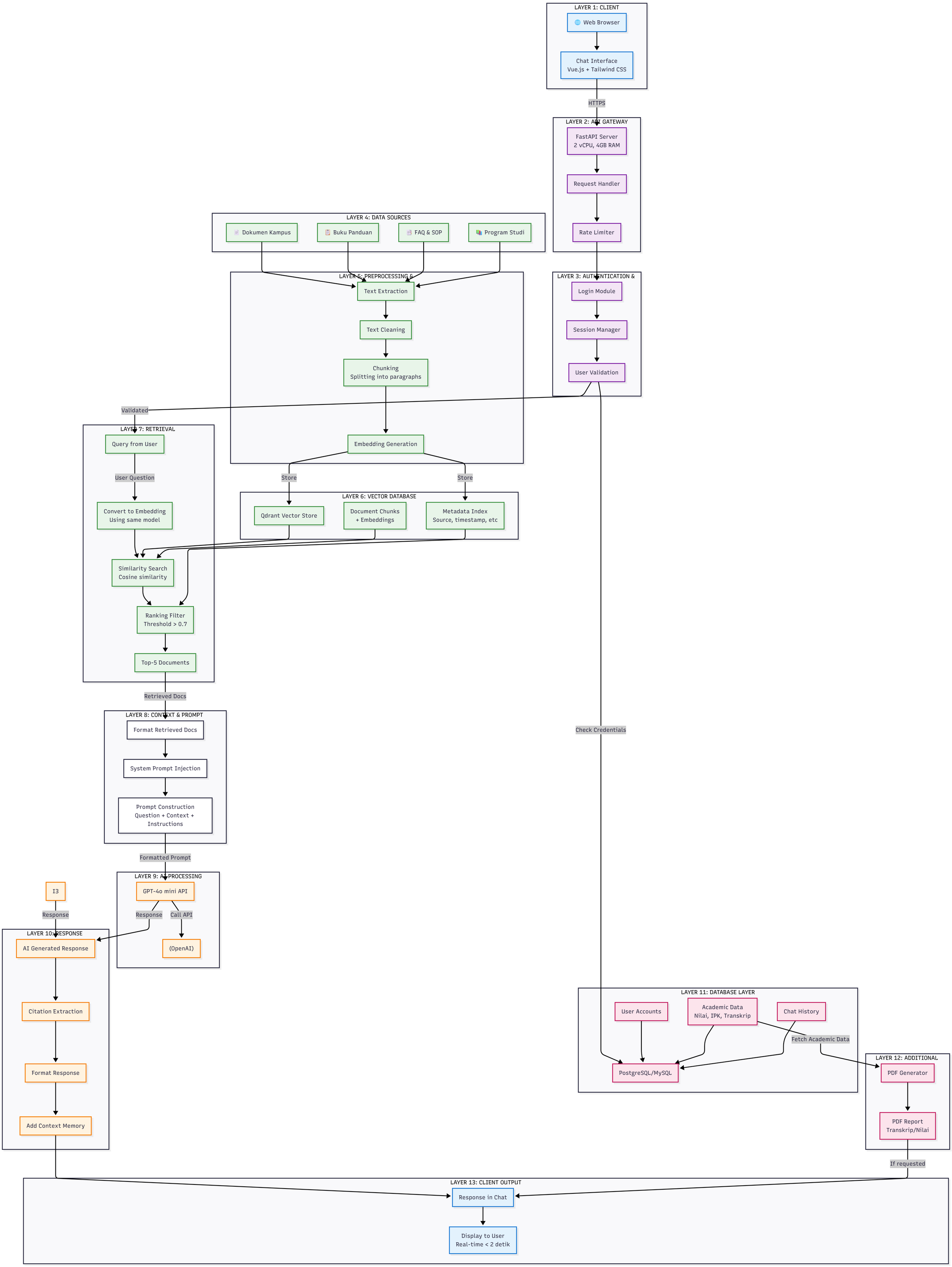
LAYER 1: CLIENT
Web Browser
Chat Interface
Vue.js + Tailwind CSS
HTTPS
LAYER 2: API GATEWAY
FastAPI Server
2 vCPU, 4GB RAM
Request Handler
Rate Limiter
LAYER 3: AUTHENTICATION &
Login Module
Session Manager
User Validation
LAYER 4: DATA SOURCES
Dokumen Kampus
Buku Panduan
FAQ & SOP
Program Studi
LAYER 5: PREPROCESSING &
Text Extraction
Text Cleaning
Chunking
Splitting into paragraphs
Embedding Generation
Validated
Store
Store
LAYER 6: VECTOR DATABASE
Qdrant Vector Store
Document Chunks
+ Embeddings
Metadata Index
Source, timestamp, etc
LAYER 7: RETRIEVAL
Query from User
User Question
Convert to Embedding
Using same model
Similarity Search
Cosine similarity
Ranking Filter
Threshold > 0.7
Top-5 Documents
Retrieved Docs
LAYER 8: CONTEXT & PROMPT
Format Retrieved Docs
System Prompt Injection
Prompt Construction
Question + Context +
Instructions
Formatted Prompt
LAYER 9: AI PROCESSING
GPT-4o mini API
Response
Call API
(OpenAI)
I3
Response
LAYER 10: RESPONSE
AI Generated Response
Citation Extraction
Format Response
Add Context Memory
Check Credentials
LAYER 11: DATABASE LAYER
User Accounts
Academic Data
Nilai, IPK, Transkrip
Chat History
Fetch Academic Data
PostgreSQL/MySQL
LAYER 12: ADDITIONAL
PDF Generator
PDF Report
Transkrip/Nilai
If requested
LAYER 13: CLIENT OUTPUT
Response in Chat
Display to User
Real-time < 2 detik